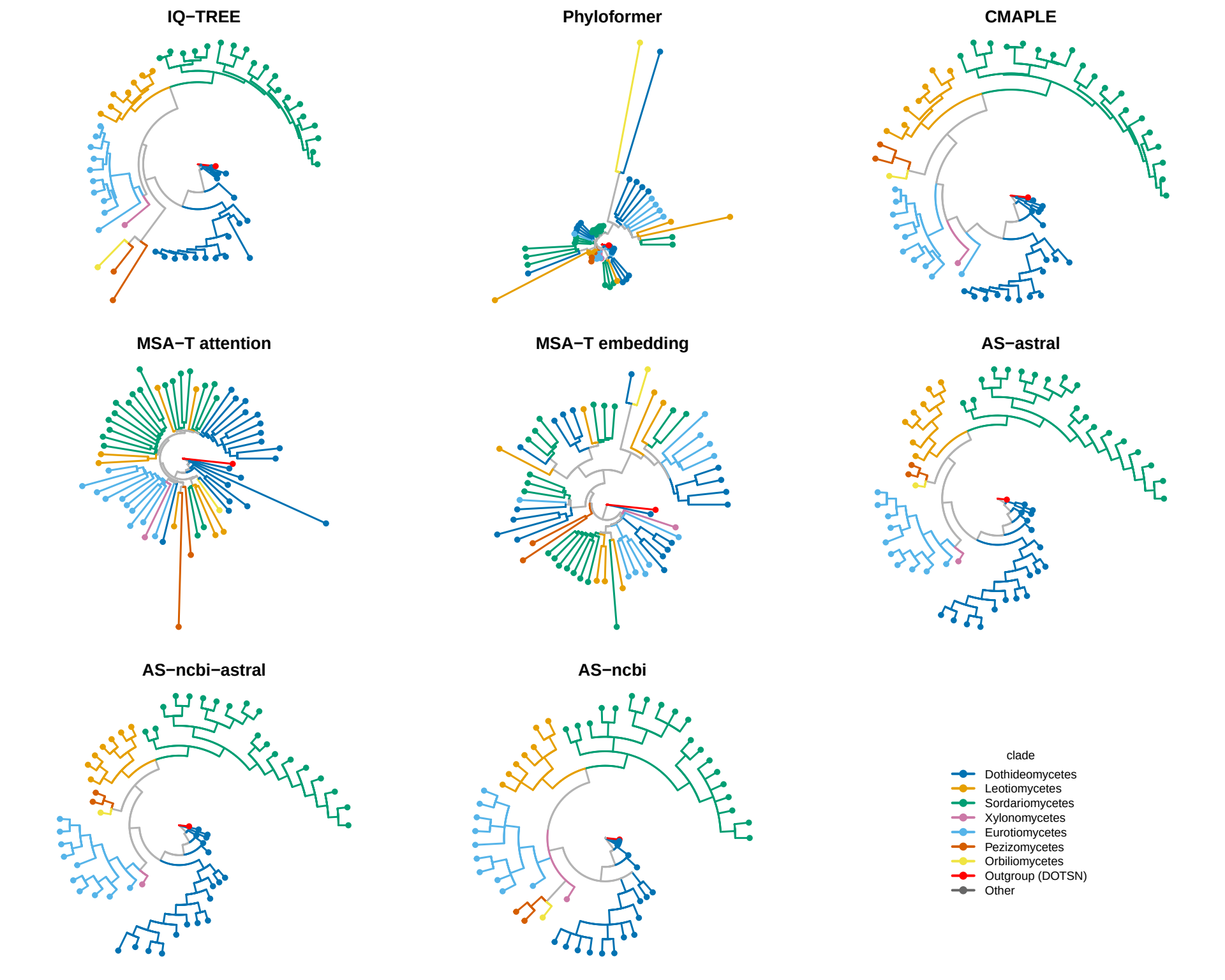
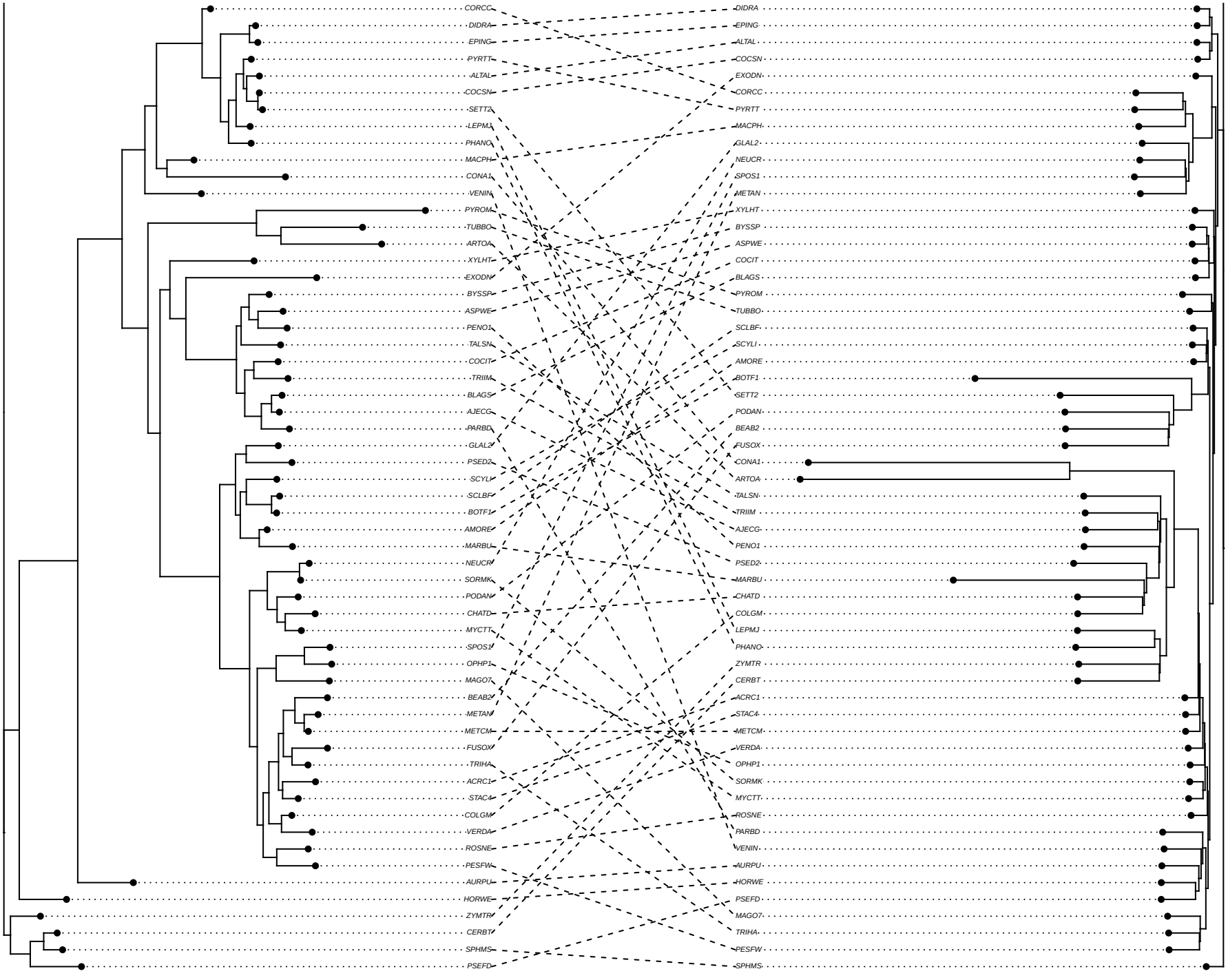
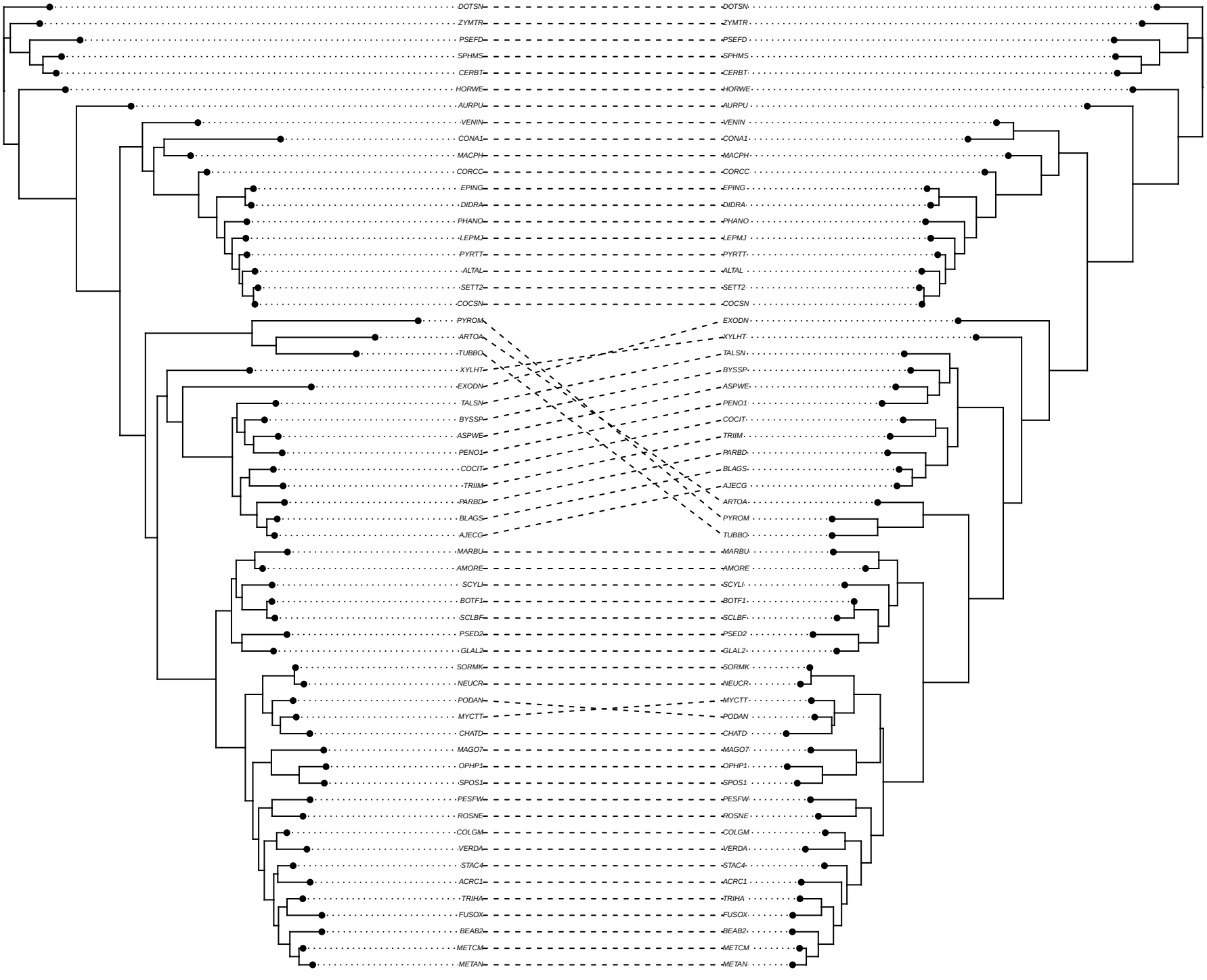
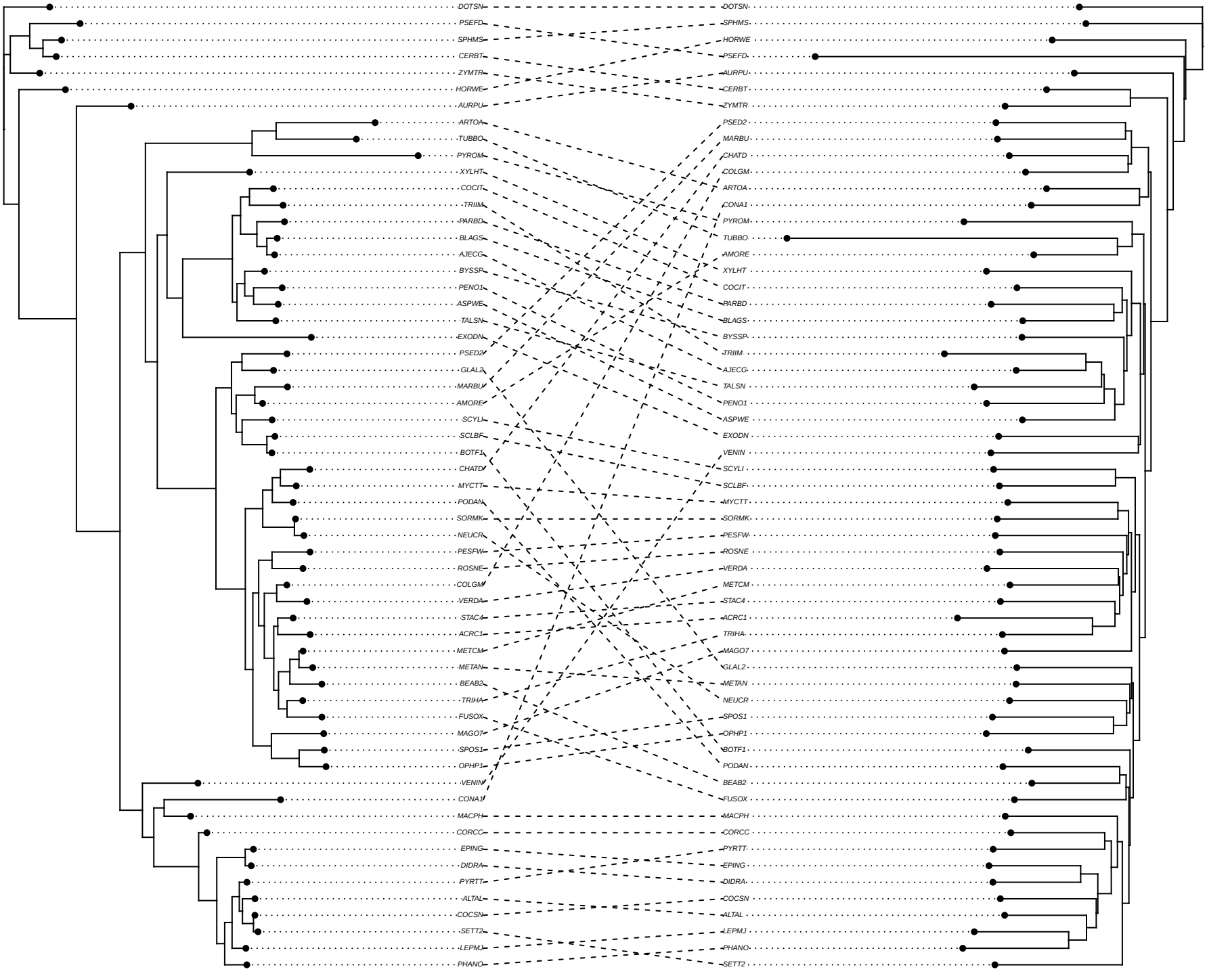
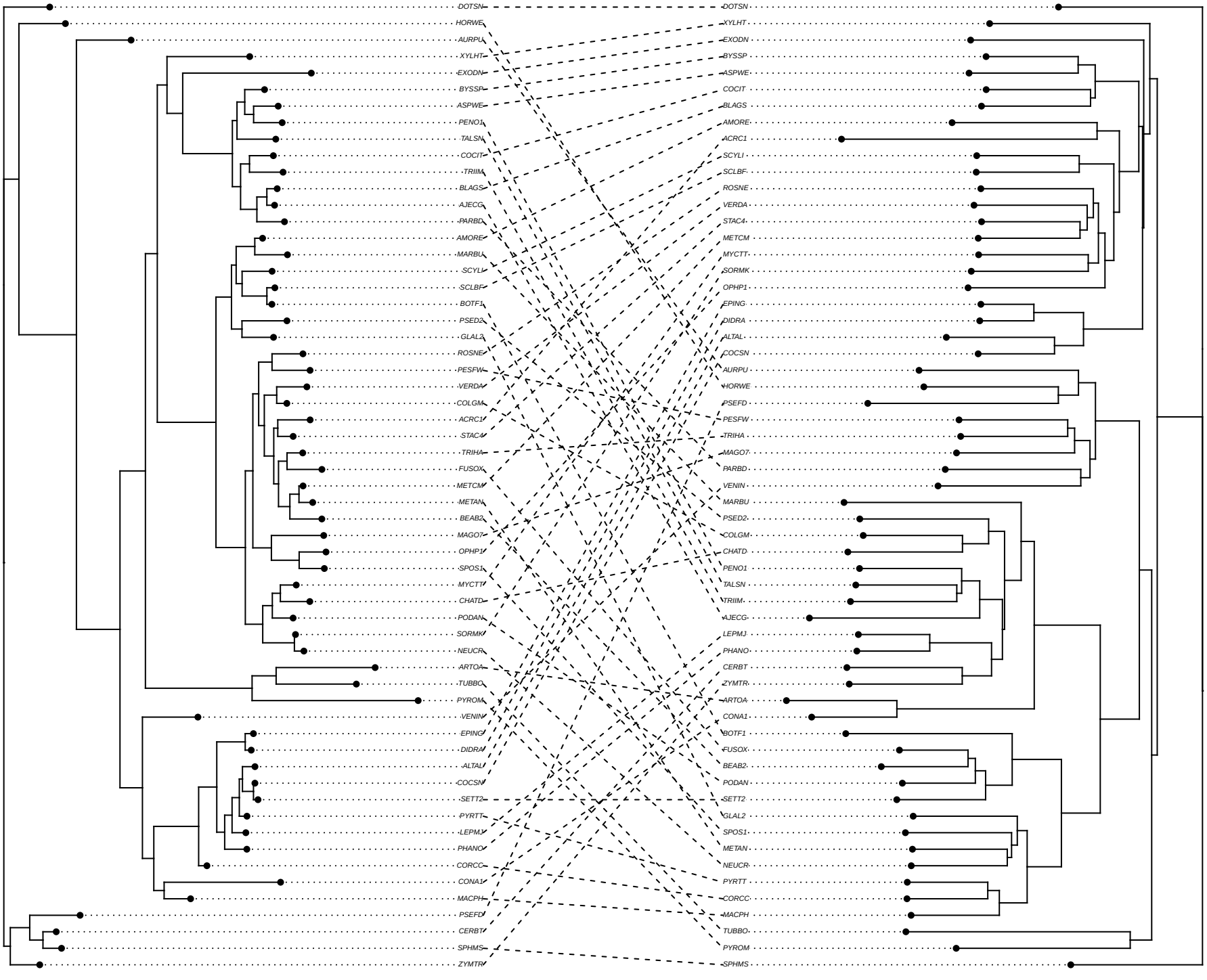

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AS−astral
AS−ncbi−astral
AS−ncbi
clade
Dothideomycetes
Leotiomycetes
Sordariomycetes
Xylonomycetes
Eurotiomycetes
Pezizomycetes
Orbiliomycetes
Outgroup (DOTSN)
Other


IQ−TREE vs Phylolformer
CORCC
DIDRA
EPING
PYRTT
ALTAL
COCSN
SETT2
LEPMJ
PHANO
MACPH
CONA1
VENIN
PYROM
TUBBO
ARTOA
XYLHT
EXODN
BYSSP
ASPWE
PENO1
TALSN
COCIT
TRIIM
BLAGS
AJECG
PARBD
GLAL2
PSED2
SCYLI
SCLBF
BOTF1
AMORE
MARBU
NEUCR
SORMK
PODAN
CHATD
MYCTT
SPOS1
OPHP1
MAGO7
BEAB2
METAN
METCM
FUSOX
TRIHA
ACRC1
STAC4
COLGM
VERDA
ROSNE
PESFW
AURPU
HORWE
ZYMTR
CERBT
SPHMS
PSEFD
DIDRA
EPING
ALTAL
COCSN
EXODN
CORCC
PYRTT
MACPH
GLAL2
NEUCR
SPOS1
METAN
XYLHT
BYSSP
ASPWE
COCIT
BLAGS
PYROM
TUBBO
SCLBF
SCYLI
AMORE
BOTF1
SETT2
PODAN
BEAB2
FUSOX
CONA1
ARTOA
TALSN
TRIIM
AJECG
PENO1
PSED2
MARBU
CHATD
COLGM
LEPMJ
PHANO
ZYMTR
CERBT
ACRC1
STAC4
METCM
VERDA
OPHP1
SORMK
MYCTT
ROSNE
PARBD
VENIN
AURPU
HORWE
PSEFD
MAGO7
TRIHA
PESFW
SPHMS

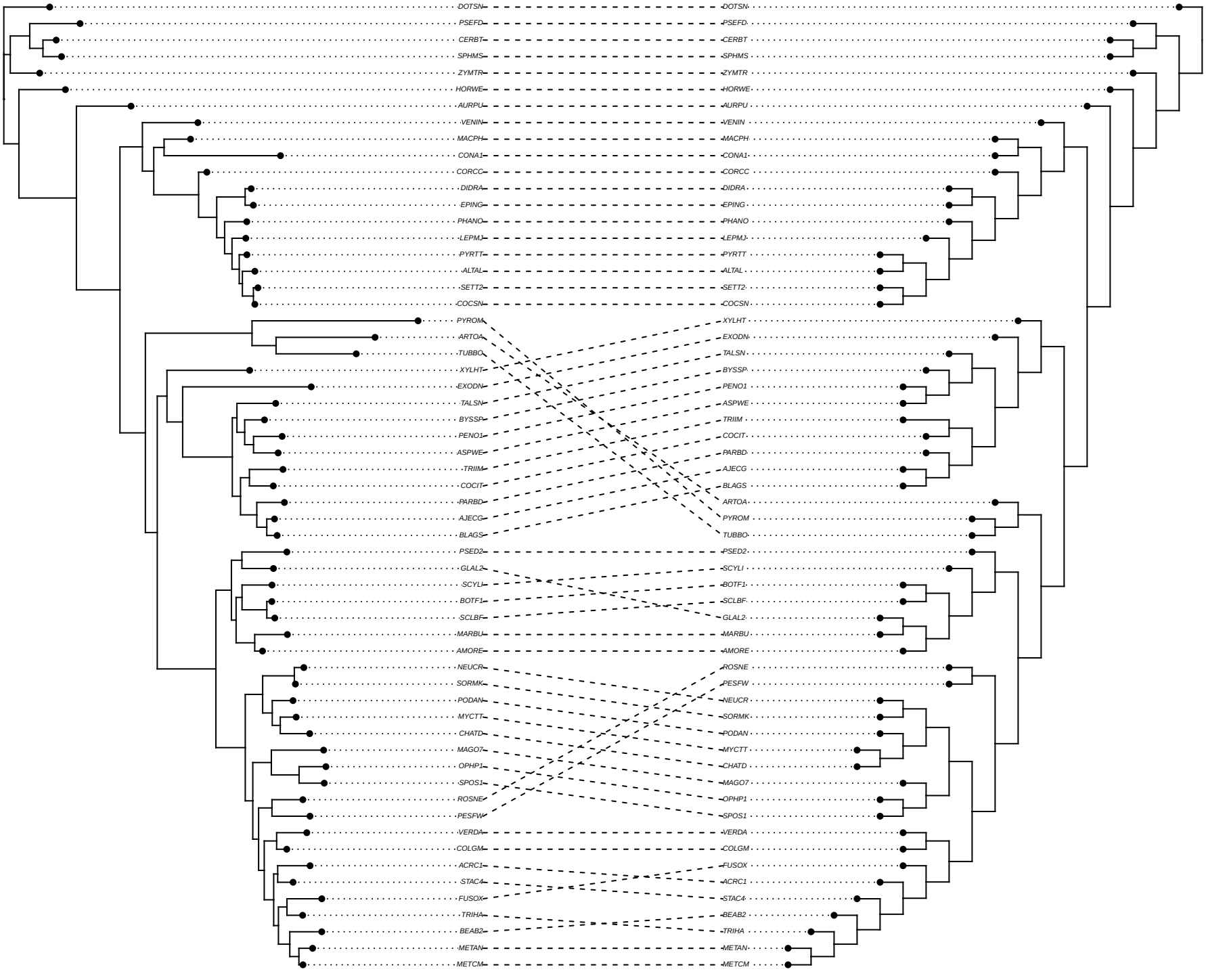
IQ-TREE vs CMAPLE



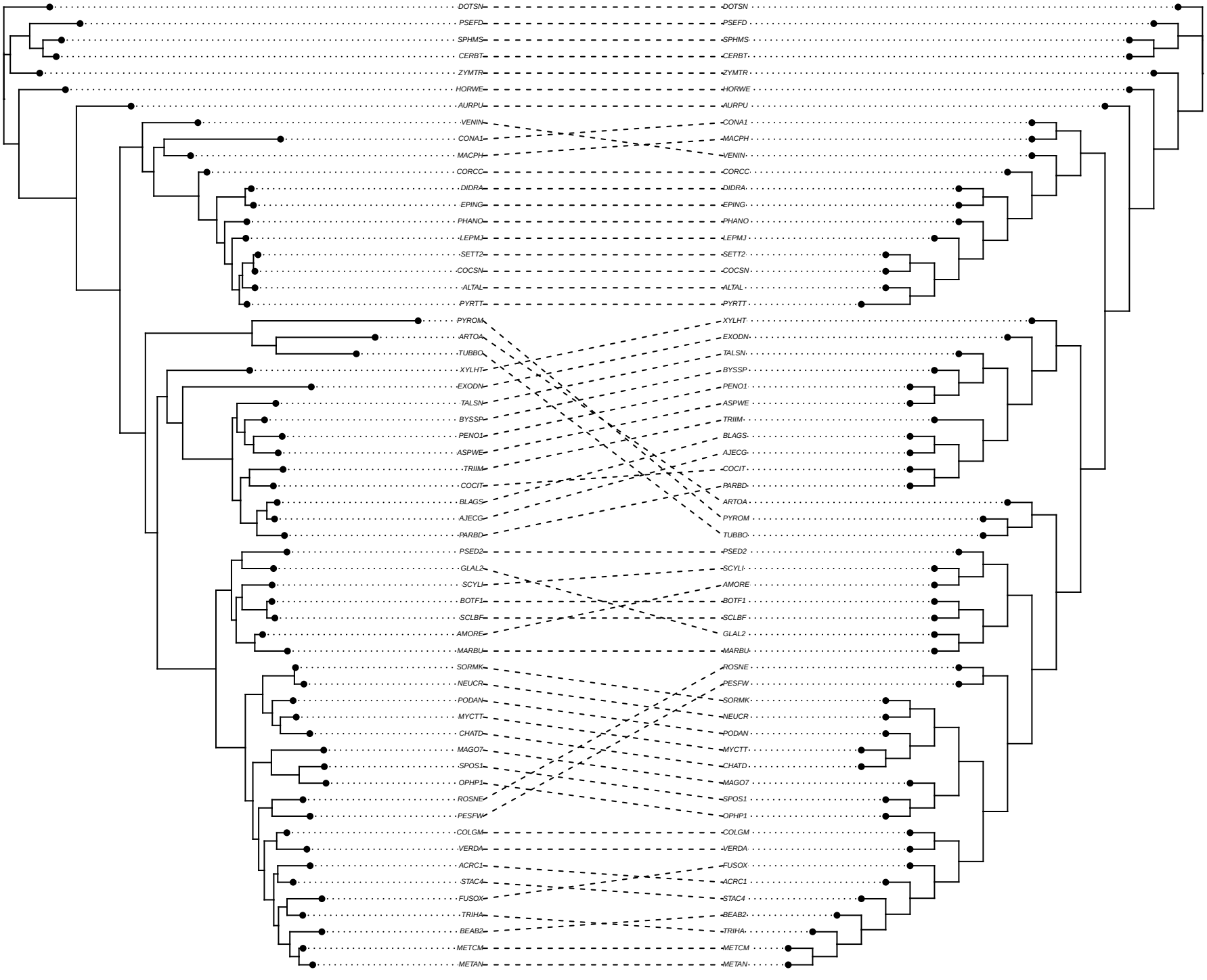

IQ−TREE vs MSA−Pattention
DOTSN
PSEFD
SPHMS
CERBT
ZYMTR
HORWE
AURPU
ARTOA
TUBBO
PYROM
XYLHT
COCIT
TRIIM
PARBD
BLAGS
AJECG
BYSSP
PENO1
ASPWE
TALSN
EXODN
PSED2
GLAL2
MARBU
AMORE
SCYLI
SCLBF
BOTF1
CHATD
MYCTT
PODAN
SORMK
NEUCR
PESFW
ROSNE
COLGM
VERDA
STAC4
ACRC1
METCM
METAN
BEAB2
TRIHA
FUSOX
MAGO7
SPOS1
OPHP1
VENIN
CONA1
MACPH
CORCC
EPING
DIDRA
PYRTT
ALTAL
COCSN
SETT2
LEPMJ
PHANO
DOTSN
SPHMS
HORWE
PSEFD
AURPU
CERBT
ZYMTR
PSED2
MARBU
CHATD
COLGM
ARTOA
CONA1
PYROM
TUBBO
AMORE
XYLHT
COCIT
PARBD
BLAGS
BYSSP
TRIIM
AJECG
TALSN
PENO1
ASPWE
EXODN
VENIN
SCYLI
SCLBF
MYCTT
SORMK
PESFW
ROSNE
VERDA
METCM
STAC4
ACRC1
TRIHA
MAGO7
GLAL2
METAN
NEUCR
SPOS1
OPHP1
BOTF1
PODAN
BEAB2
FUSOX
MACPH
CORCC
PYRTT
EPING
DIDRA
COCSN
ALTAL
LEPMJ
PHANO
SETT2


IQ−TREE vs MSA−T embedding
DOTSN
HORWE
AURPU
XYLHT
EXODN
BYSSP
ASPWE
PENO1
TALSN
COCIT
TRIIM
BLAGS
AJECG
PARBD
AMORE
MARBU
SCYLI
SCLBF
BOTF1
PSED2
GLAL2
ROSNE
PESFW
VERDA
COLGM
ACRC1
STAC4
TRIHA
FUSOX
METCM
METAN
BEAB2
MAGO7
OPHP1
SPOS1
MYCTT
CHATD
PODAN
SORMK
NEUCR
ARTOA
TUBBO
PYROM
VENIN
EPING
DIDRA
ALTAL
COCSN
SETT2
PYRTT
LEPMJ
PHANO
CORCC
CONA1
MACPH
PSEFD
CERBT
SPHMS
ZYMTR
DOTSN
XYLHT
EXODN
BYSSP
ASPWE
COCIT
BLAGS
AMORE
ACRC1
SCYLI
SCLBF
ROSNE
VERDA
STAC4
METCM
MYCTT
SORMK
OPHP1
EPING
DIDRA
ALTAL
COCSN
AURPU
HORWE
PSEFD
PESFW
TRIHA
MAGO7
PARBD
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MARBU
PSED2
COLGM
CHATD
PENO1
TALSN
TRIIM
AJECG
LEPMJ
PHANO
CERBT
ZYMTR
ARTOA
CONA1
BOTF1
FUSOX
BEAB2
PODAN
SETT2
GLAL2
SPOS1
METAN
NEUCR
PYRTT
CORCC
MACPH
TUBBO
PYROM
SPHMS

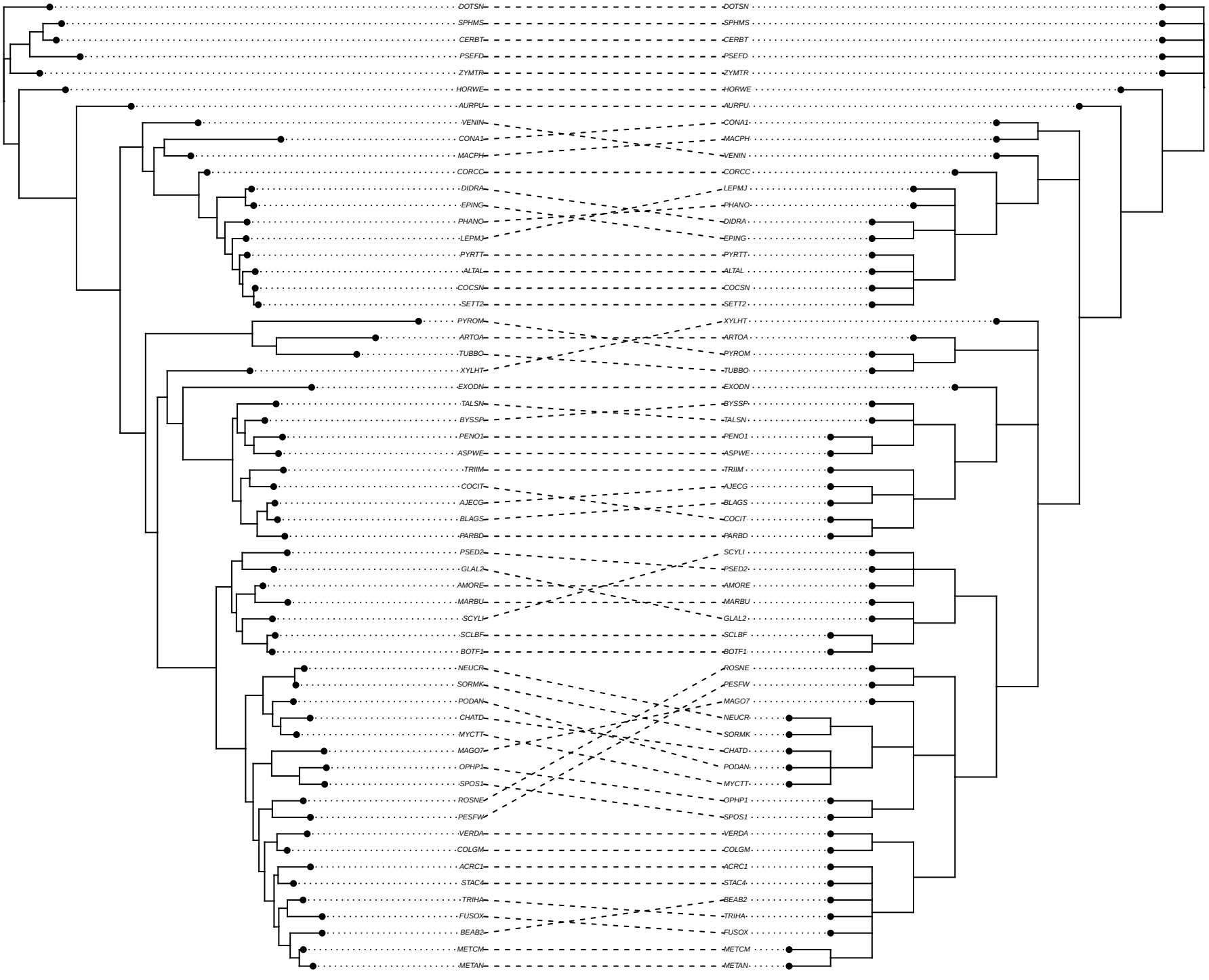
IQ−TREE vs AS−astral

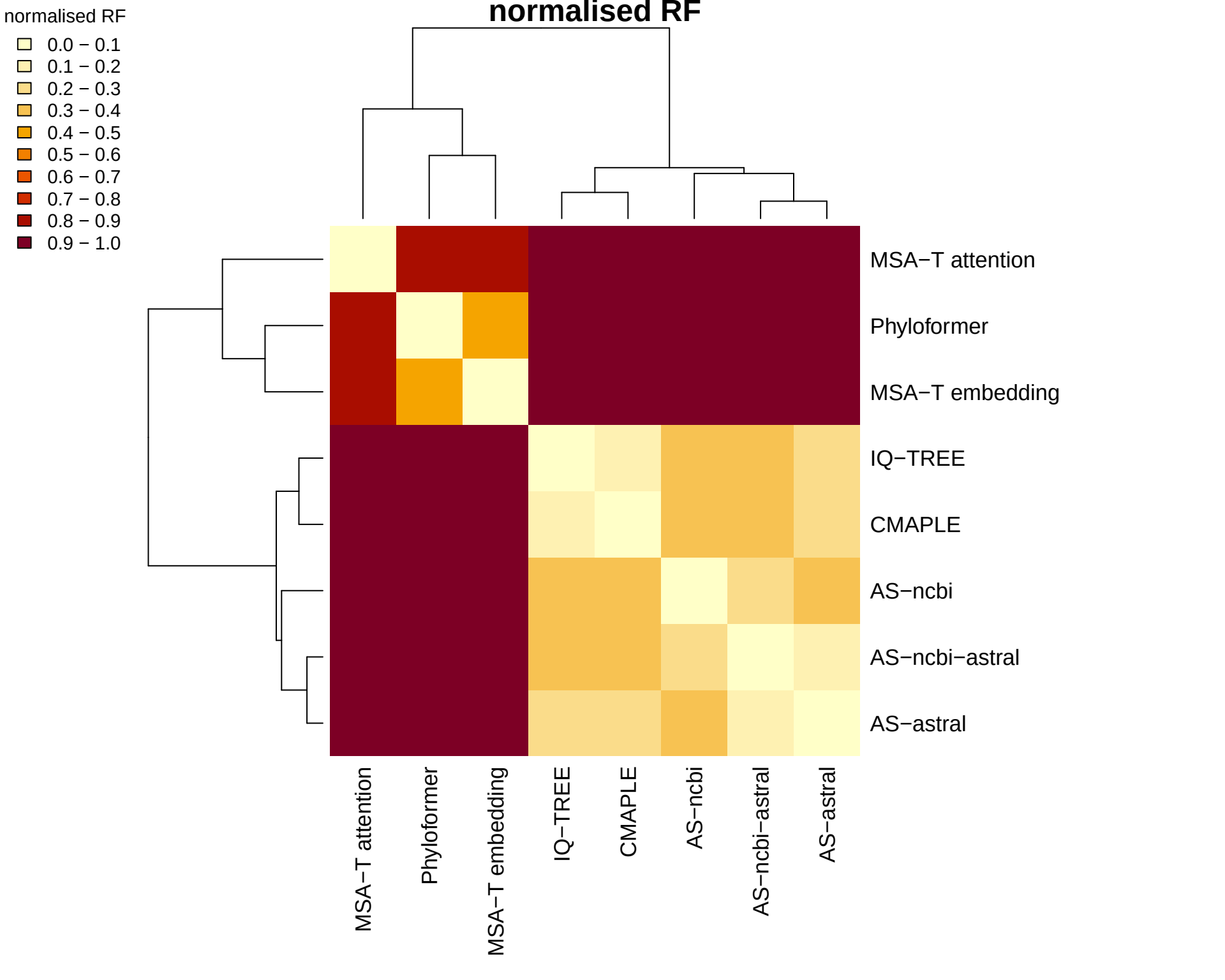


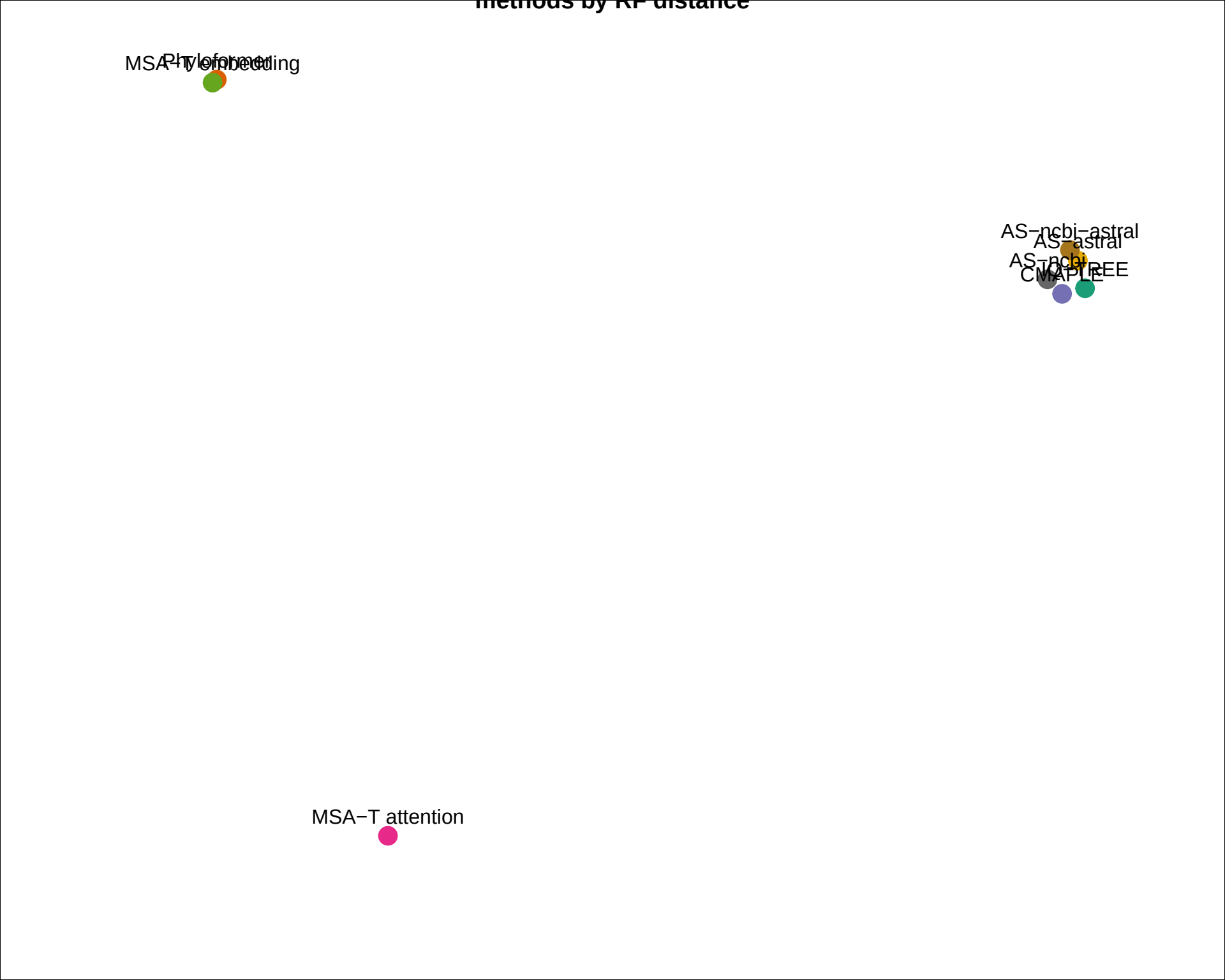
IQ-TREE vs AS-ncbi-astral

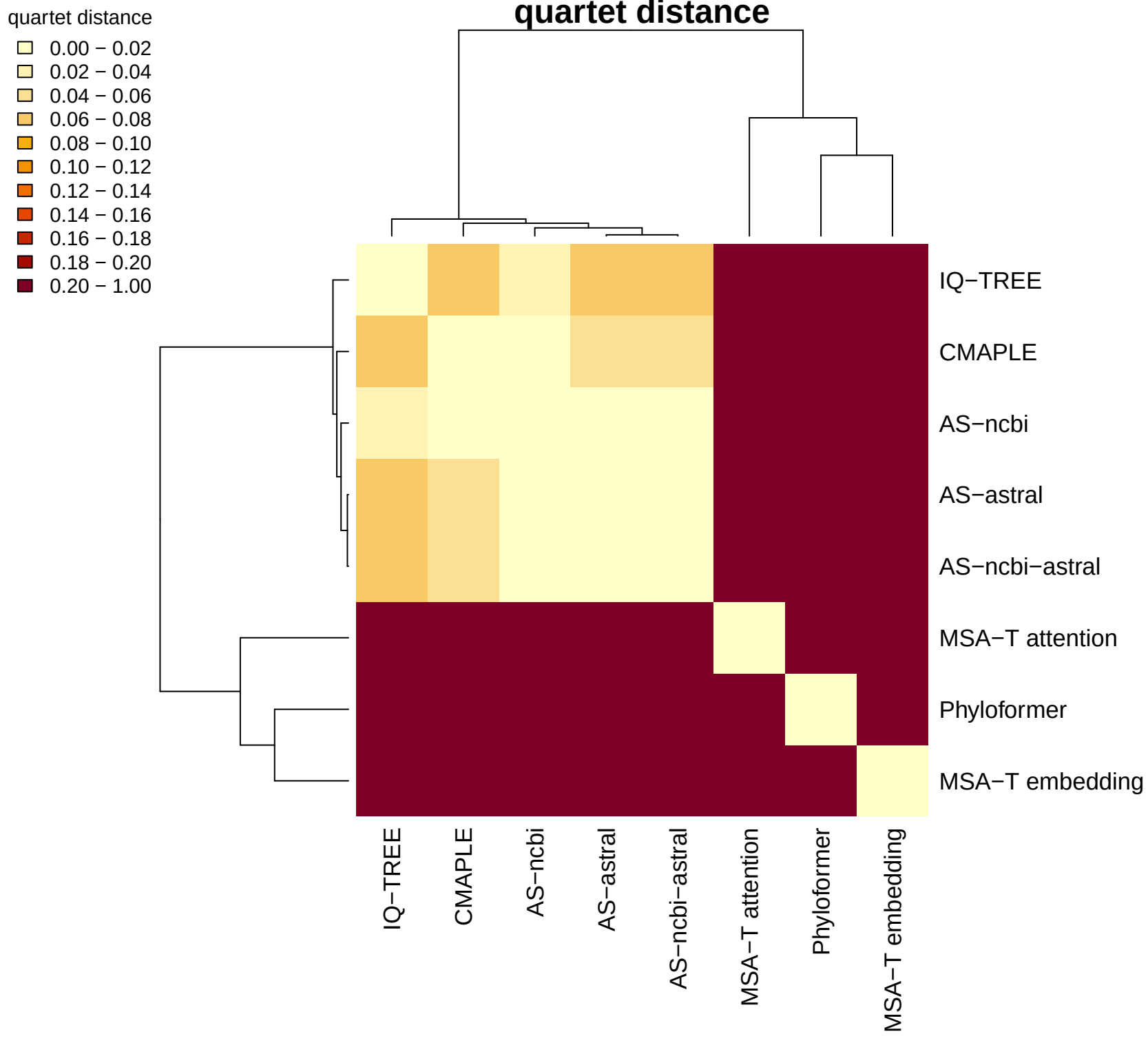


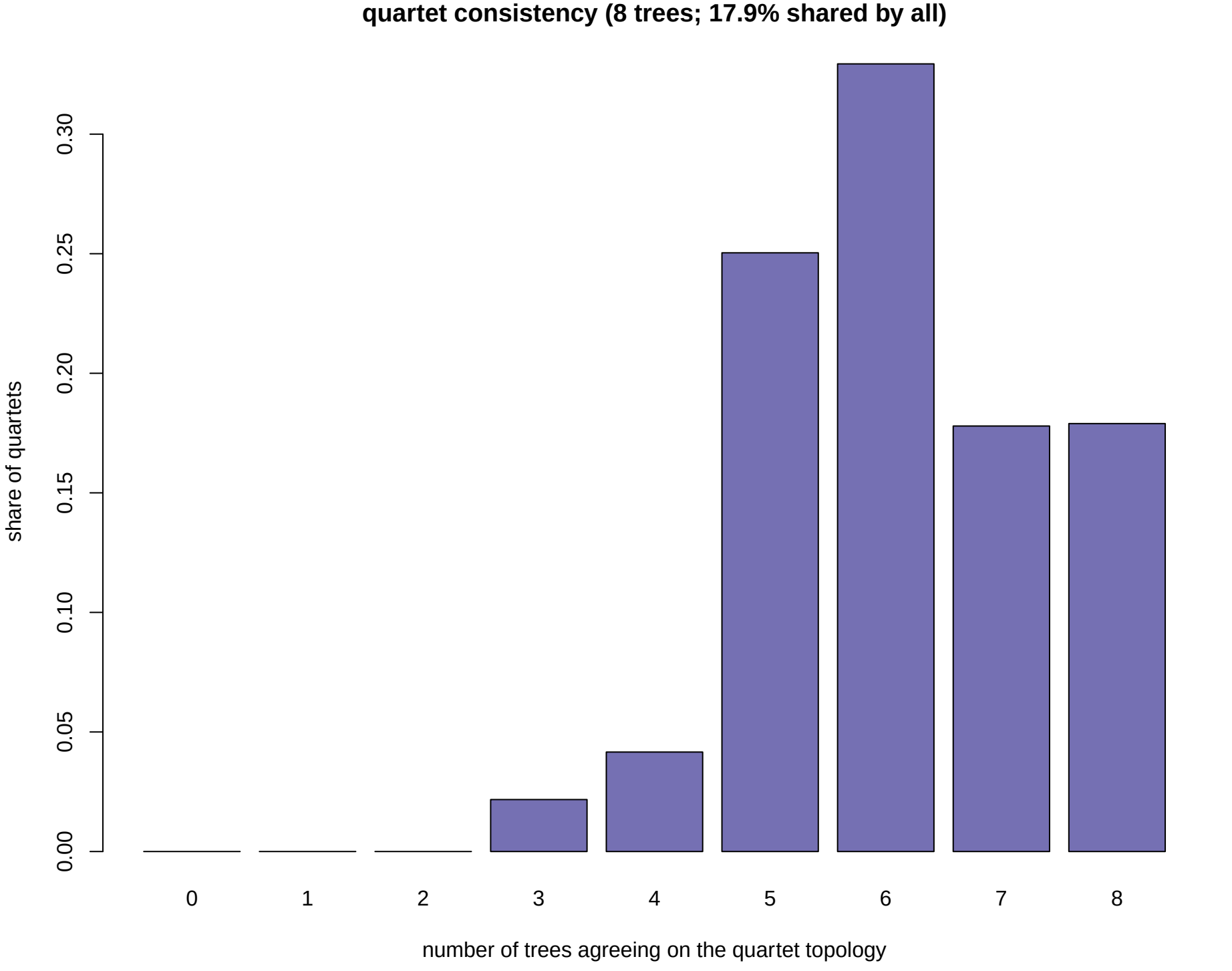
IQ-TREE vs AS-ncbi

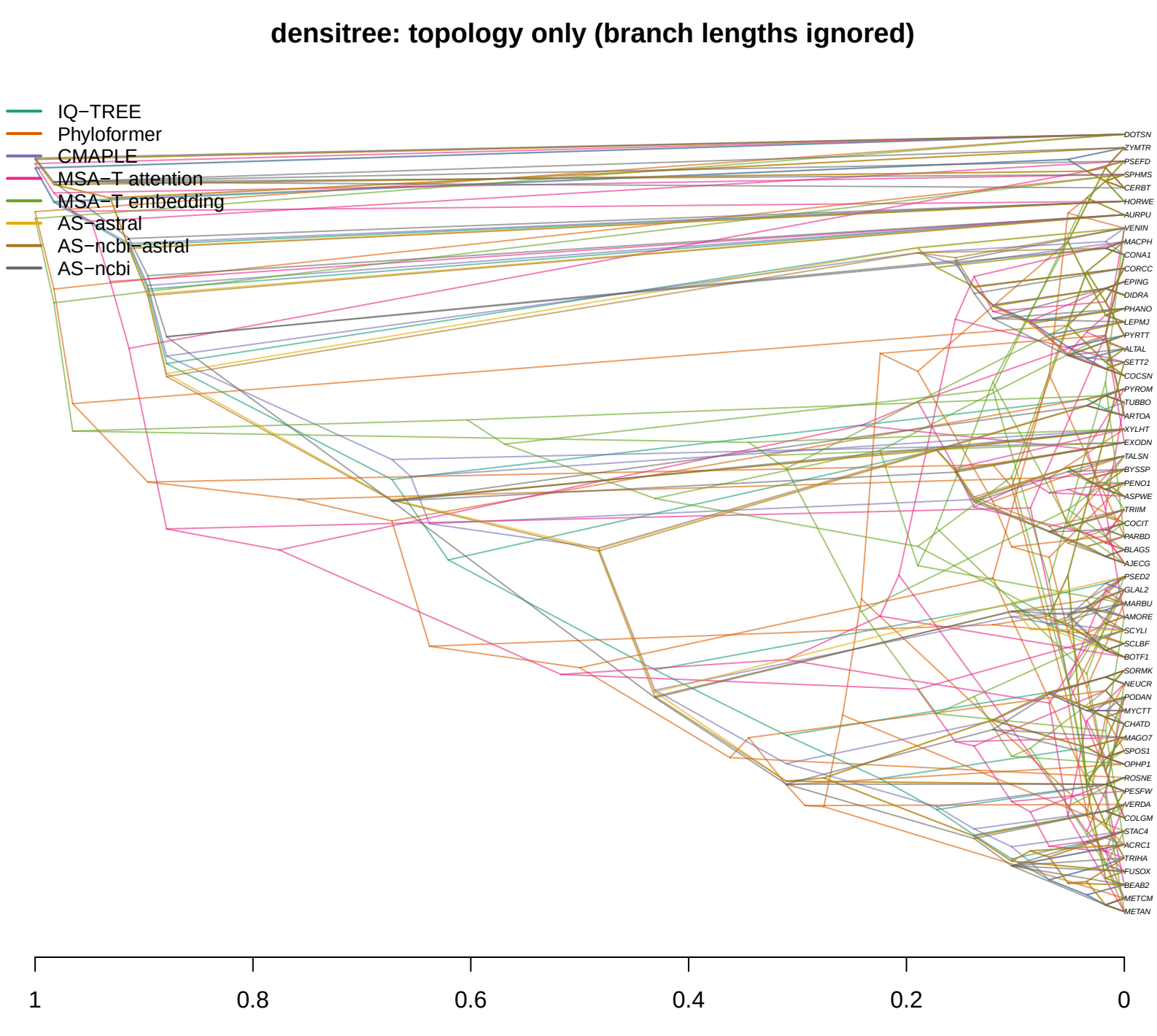


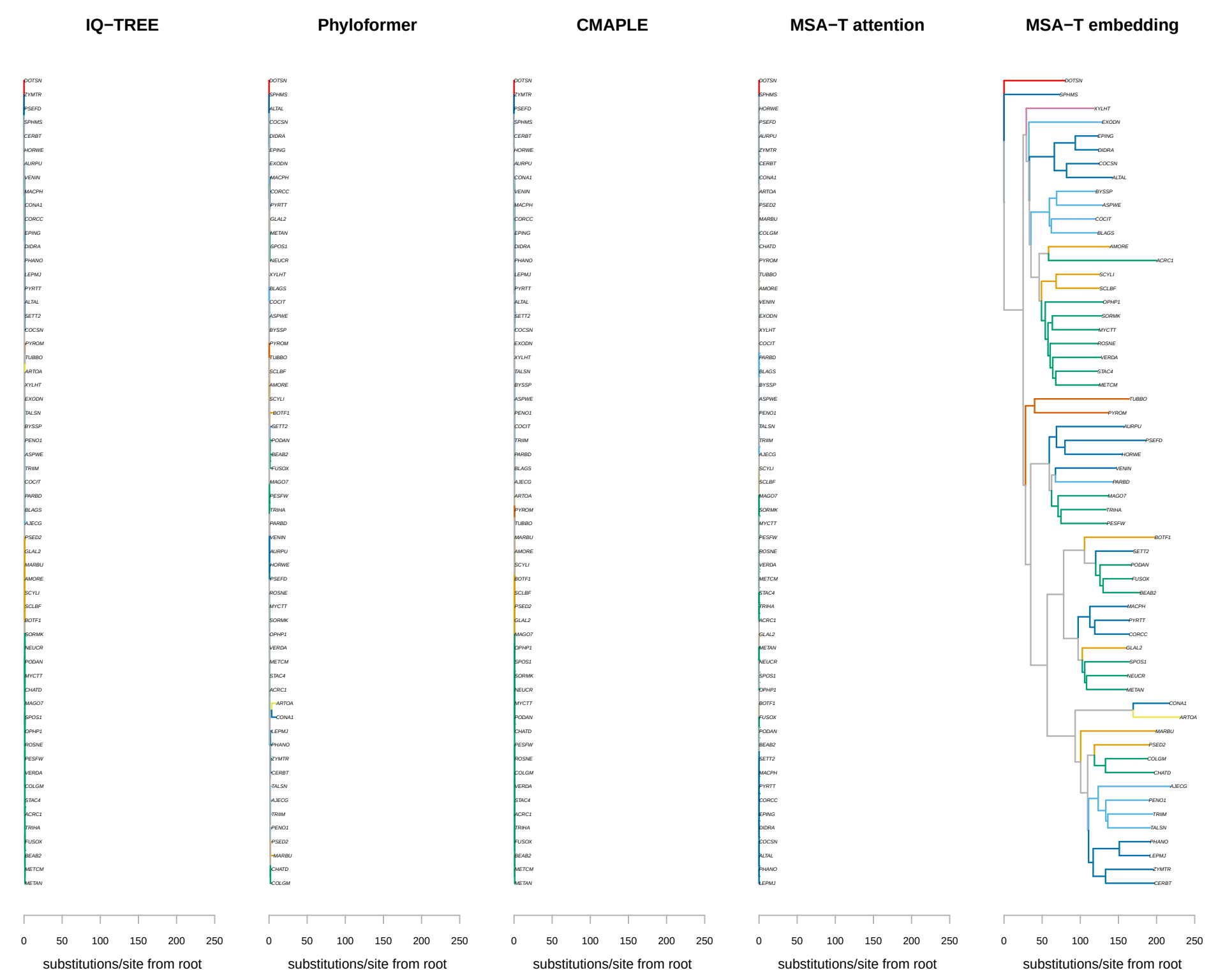

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
AS−ncbi
AS−ncbi−astral
AS−astral
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
AS−ncbi
AS−ncbi−astral
AS−astral


methods by RF distance
MSA−T embedding
Phyloformer
AS−ncbi−astral
AS−astral
AS−ncbi
CMAPLE
IQ−TREE
MSA−T attention

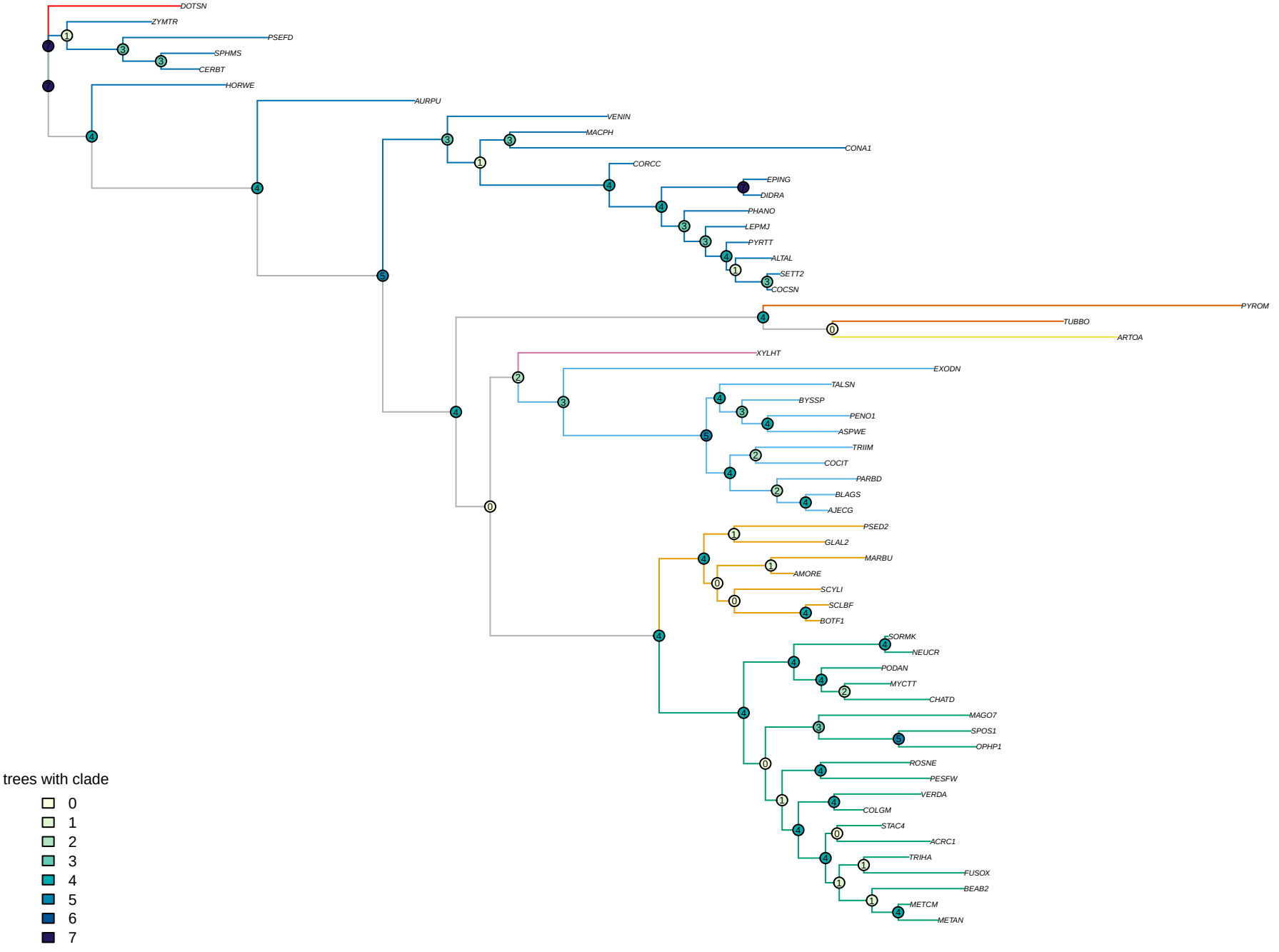

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
IQ−TREE
CMAPLE
AS−ncbi
AS−astral
AS−ncbi−astral
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
AS−ncbi
AS−astral
AS−ncbi−astral
MSA−T attention
Phyloformer
MSA−T embedding


quartet consistency (8 trees; 17.9% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

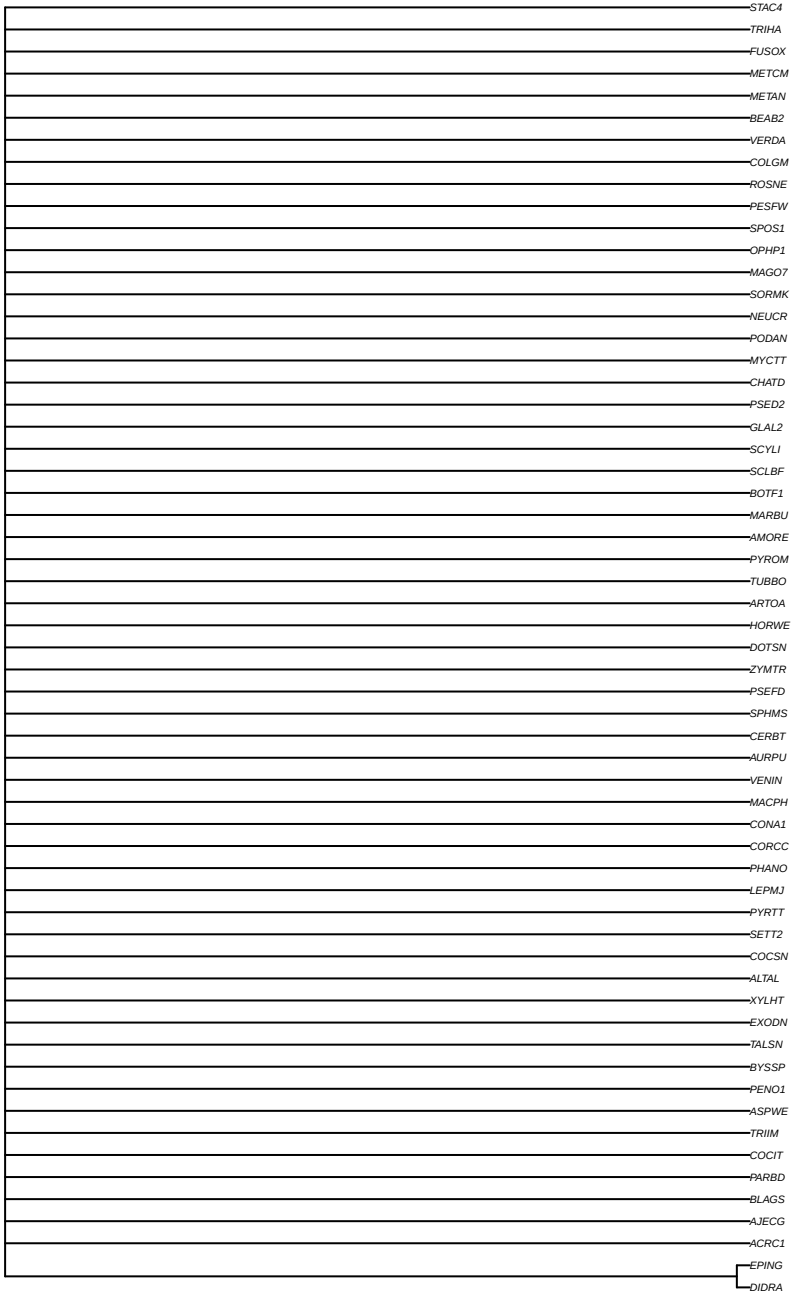

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AS−astral
AS−ncbi−astral
AS−ncbi
DOTSN
ZYMTR
PSEFD
SPHMS
CERBT
HORWE
AURPU
VENIN
MACPH
CONA1
CORCC
EPING
DIDRA
PHANO
LEPMJ
PYRTT
ALTAL
SETT2
COCSN
PYROM
TUBBO
ARTOA
XYLHT
EXODN
TALSN
BYSSP
PENO1
ASPWE
TRIIM
COCIT
PARBD
BLAGS
AJECG
PSED2
GLAL2
MARBU
AMORE
SCYLI
SCLBF
BOTF1
SORMK
NEUCR
PODAN
MYCTT
CHATD
MAGO7
SPOS1
OPHP1
ROSNE
PESFW
VERDA
COLGM
STAC4
ACRC1
TRIHA
FUSOX
BEAB2
METCM
METAN
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
DOTSN
ZYMTR
PSEFD
SPHMS
CERBT
HORWE
AURPU
VENIN
MACPH
CONA1
CORCC
EPING
DIDRA
PHANO
LEPMJ
PYRTT
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AMORE
SCYLI
SCLBF
BOTF1
SORMK
NEUCR
PODAN
MYCTT
CHATD
MAGO7
SPOS1
OPHP1
ROSNE
PESFW
VERDA
COLGM
STAC4
ACRC1
TRIHA
FUSOX
BEAB2
METCM
METAN
0
50
100
150
200
250
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

